SUMMARY OF NON-DETRIMENT FINDINGS OF

AGAR TREES (AQULARIA MALACCENSIS)

1. **Introduction**:

   a. MoEF&CC commissioned the NDF Study through the Botanical Survey of India (BSI) with a Budget Rs 30.60 lakh and Project Tenure of 4 months.
   b. BSI had submitted the report in February 2024. The work was carried out by following the 'CITES Non-detriment Findings Guidance for Perennial Plants' (Wolf & al 2016).
   c. Ensuring trade within sustainable limits is at the core of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). According to the Convention, Parties shall allow trade in specimens of species included in Appendix II only if the Scientific Authority of the State of export has advised that "such export will not be detrimental to the survival of that species" (Article IV).
   d. Aquilaria malaccensis is listed as a Critically Endangered (CR) Species in the IUCN Red List of Threatened Species. The species is listed under CITES Appendix-II (w.e.f. 16th February 1995).
   e. Major Agarwood markets in Hojai (Assam) and Calicut (Kerala) were surveyed.
   f. Several Agarwood Processing units and other Agarwood-based units were visited in Assam, Karnataka, Kerala, Manipur, Nagaland and Tripura.

2. **Cultivated Agar Tree National Population estimation (>2 years of age & >10 cm GBH**:

Sl	Name of the State	Estimated number (in lakh)	Percentage
1	**Assam**	**1143.7**	**82.38**
2	Tripura	151.5	10.81
3	Manipur	32	2.29
4	Karnataka	24	1.71
5	Nagaland	23	1.64
6	Andhra Pradesh	10	<1
7	Kerala	10	<1
8	Maharashtra	3	<1
9	Meghalaya	1.72	<1
10	Arunachal Pradesh	0.25	<1
11	Telangana	0.20	<1
12	Mizoram	0.10	<1
13	Tamil Nadu	0.10	<1
14	West Bengal	0.06	<1
15	Others	0.03	<1
Total		**1398.9 lakh**	
		139.89 million	
		13.989 Crore	

3. **Cultivated Agar Tree Assam Population estimation (>2 years of age & >10 cm GBH**:

Sl	Name of the District	Estimated number (in lakh)	Percentage
1	Golaghat	450	39.3
2	Jorhat	200	17.5
3	Sivasagar	175	15.3
4	Karimganj	150	13.11
5	Cachar	100	8.74
6	Charaideo	50	4.37
7	Dibrugarh	40	3.5
8	Hojai	2	<1
9	Nagaon	1.2	<1
10	Dima Hasao	1	<1
11	Hailakandi	0.17	<1
12	Biswanath	0.1	<1
13	West Karbi Anglong	0.1	<1
14	Goalpara	0.06	<1
15	Tinsukia	0.03	<1
16	Dhemaji	0.01	<1
Total		**1143.7 Lakh**	

4. **Table showing proposed State-wise Quota for export of Agarwood products**:

Sl	Name of the State	Proposed Export Quota (in Kg)	
		Agarwood Chips/ Powder/Saw-Dust	Agar Oil
1	Assam	119400	5560
2	Tripura	22500	1047
3	Manipur	4000	200
4	Nagaland	3400	180
5	Karnataka	1200	35
6	Meghalaya	200	9
7	Kerala	80	4
8	Others	300	15

5. **Agarwood Product international valuation data**:

Sl	Product	Value per Kg (US $)	
		Low Quality	High Quality
1	Agarwood (Per Kg)	100	1,00,000
2	Oud Oil (Per Ltr)	-	80,000

6. **Agarwood Product valuation data in Assam & Tripura**:

Sl	Product	Value per Kg (INR)	
		Low Quality	**High Quality**
1	Chips (Per Kg)	10,000	3,00,000
2	Agar Oil (Per tola)	6,000	17,500
3	Sem (Highest quality of agarwood Per Kg)	2,00,000	Several Crores
4	Boya Oil (Per tola)	1,000	1,500
5	Bangtang (Per Kg)	1500	3500
6	Chap Bangtang (Per Kg)	30,000	50,000

7. **Magnitude of the trade**:

   a. The global agarwood chips market expanded at 7.7% Compound Annual Growth Rate (CAGR) during 2018 to 2022 and ended up at a valuation of US$ 44,050.7 million (Rs. 3,52,405.6 Crore approx.) in 2023.
   b. The global sales of agarwood chips are estimated to increase at 7.1% Compound Annual Growth Rate (CAGR) to reach a market size of US$ 86,467.6 million (Rs. 6,91,740.8 Crore approx.) by the end of 2023.
   c. The global market for agarwood is estimated at US$ 32 billion and by the end of 2029, it is expected to reach US$64 billion. (Rs. 5,12,000 Crore approx.)
   d. According to CITES Trade Database:
      i. India imported more than 142 tonnes of agarwood from 2017 to 2021.
      ii. India re-exported over 141 tonnes of agarwood from 2017 to 2021.
   e. According to WCCB Database:
      i. Agarwood import from Eastern Region of India from 2014 to 2022: 161 consignments – 4.6868 lakh kg. (74% of total import in India)
      ii. Agarwood re-export from Eastern Region of India from 2012 to 2022: 429 consignments – 3.48275 lakh kg. (92% of total import in India)